(1)

Last time: R/I and R/J are fields of finite size p and q, respectively $(I \neq J)$. $\Rightarrow$ Taking power in $R/I\cap J$ is an operation $x \mapsto x^e$, in which the exponent $e \in \mathbb{N}$ only matters modulo $(p-1)(q-1)$. This means $x^{e+m(p-1)(q-1)} = x^e$ holds for every $x \in R/I\cap J$ and every $m \in \mathbb{Z}_{\geq 0}$. $\leftarrow$ Summary of $\boxed{1.9.2}$ and $\boxed{1.9.1}$

So, if $\gcd(e,(p-1)(q-1)) = 1$ and you've determined $d \in \mathbb{N}$ with $ed \equiv 1 \bmod (p-1)(q-1)$, then the power functions $x \mapsto x^e$ and $x \mapsto x^d$ on $R/I\cap J$ are inverse to each other.

$\boxed{1.3}$ The Rivest-Shamir-Adleman Cryptosystem (RSA)

$\boxed{\text{ssh}}$ $\rightarrow$ RSA, $\rightarrow$ Elliptic curves.

We do as in 1.9.2 with $R = \mathbb{Z}$, $I = p\mathbb{Z}$, $J = q\mathbb{Z}$ with $p \neq q$ and p, q being prime.

Then $R/I\cap J = \mathbb{Z}/p\mathbb{Z}\cap q\mathbb{Z} = \mathbb{Z}/pq\mathbb{Z}$ $\leftarrow$ our messages and ciphertexts come from this domain.

The public key is a pair (e, pq) where $e \in \mathbb{N}$ and $\gcd(e,(p-1)(q-1)) = 1$.

The sender encrypts $x \in \mathbb{Z}/pq\mathbb{Z}$ as x^e and sends x^e to the receiver.

The receiver knows e, p and q and it calculates $d \in \mathbb{N}$ such that $ed \equiv 1 \bmod (p-1)(q-1)$ (with the EEA)

The receiver takes x^e raises it to the power d and gets

$$(x^e)^d = x^{ed} = x^{ed + m(p-1)(q-1)} \quad \text{where } m \in \mathbb{N}$$

which is exactly x (the plain text)

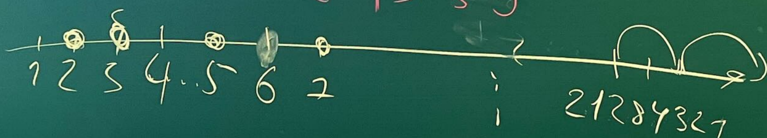
2

To break the cryptosystem, the interceptor may try factorizing pq into the factors p and q. Once this is achieved, the interceptor knows what the receiver knows (and he can calculate the d). However, no efficient techniques for factorization of integers into prime factors are known

123	✗
125	✗
127	✓

$$21284321 = 31 \cdot 109 \cdot 6299$$

$$21284339$$



$\mathbb{Z}/p\mathbb{Z}$ $\quad x \in \mathbb{Z}/p\mathbb{Z}$

$x^p = x$

4

Some of the objectives: We want to be able to deal with algebraic equations algorithmically. Theoretical complement: understand the structure of ideals in $k[x_1,\dots,x_n]$ (we've done $\mathbb{Z}$ and $k[x]$ so far, but they are special)

2.1. Monomial orders

2.1.1. Def. A binary relation $\precsim$ on a set M is called a *total order* if the following hold:

- $x \precsim y,\ y \precsim x \implies x = y$
- $x \precsim y,\ y \precsim z \implies x \precsim z$
- $x \precsim y$ or $y \precsim x$

for all $x, y, z \in M$.

$x^2 y$

$x y^2$

$x^2 y^2$

For such $\precsim$ we define the respective strict total orders by

$$x \prec y \quad :\iff \quad x \precsim y,\ x \neq y.$$

Clearly: $x \precsim y \iff x \prec y$ or $x = y$.

(so, it's enough to define $\precsim$ or $\prec$ and the other ~~other~~ one is fixed uniquely).

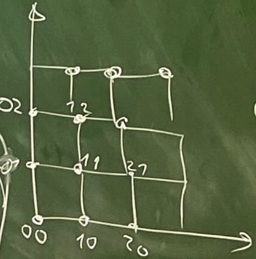
On the monomials in $k[x]$, we have the natural total order $x^0 \prec x^1 \prec x^2 \prec \dots$ derived from the strict order $<$ on $\mathbb{Z}_{\geq 0}$. In this respect $LT(f)$ for $f \in k[x]$ is the term to the largest monomial with respect to the order $\prec$ on the monomials of $k[x]$. The long division relies on $LT(f)$. The question this: how do we order monomials of $k[x_1, \dots, x_n]$. They form the set $\{x^\alpha : \alpha \in \mathbb{Z}^n_{\geq 0}\}$. x^α is given by $\alpha \in \mathbb{Z}^n_{\geq 0}$ (the exponent vector). So, we need to order $\mathbb{Z}^n_{\geq 0}$.

$$f = 2 - 3x + 4x^3 = 2 \cdot x^0 - 3 \cdot x^1 + 4 \cdot x^3$$

$$x^0 \prec x^1 \prec x^3$$

$$\boxed{0 < 1 < 3}$$

$$\begin{array}{ccccc} \vdots & & & & \\ y^2 & - & xy^2 & - & x^2y^2 \\ | & & | & & | \\ y & - & xy & - & x^2y \\ | & & | & & | \\ 1 & - & x & - & x^2 \dots \end{array}$$



There is more than one way.

(6)

2.1.2. Def. A monomial order on $\mathbb{Z}_{\geq 0}^n$ is a relation $\succ$ such that the following hold:

(i) $\succ$ is a strict total order on $\mathbb{Z}_{\geq 0}^n$

(ii) $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$, $\alpha \succ \beta$, $\gamma \in \mathbb{Z}_{\geq 0}^n \Rightarrow \alpha + \gamma \succ \beta + \gamma$.

(iii) For every nonempty subset A of $\mathbb{Z}_{\geq 0}^n$, the A has the minimal element, that is some $\gamma \in A$ satisfies $\gamma \succ \alpha$ for all $\alpha \in A \setminus \{\gamma\}$.